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**The Causal Interpretation of Two-Stage Least
Squares with Multiple Instrumental Variables**

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Contents

1	Introduction	2
1.1	Literature Review	3
2	Methodology Overview	5
2.1	Framework and Assumptions	5
2.1.1	Setup and Potential Outcomes	5
2.1.2	2SLS Estimand	5
2.2	Imbens and Angrist Monotonicity (IAM)	6
2.3	Partial Monotonicity	7
2.3.1	Response Groups under Partial Monotonicity	7
2.4	2SLS Estimand under Partial Monotonicity	9
2.4.1	Instrumental Variable Assumptions	9
2.4.2	Causal Interpretation of 2SLS under Partial Monotonicity	9
3	Simulation Study	13
4	Discussion	13
4.1	Summary	13
	References	14

1 Introduction

In many empirical applications, researchers employ multiple instrumental variables in **two-stage least squares (2SLS)** estimation in order to improve statistical efficiency. When treatment effects are constant, combining several valid instruments strengthens the first stage, reduces sampling variance, and leads to more precise estimates. In such settings, the use of multiple instruments is uncontroversial and often desirable, and efficiency considerations alone provide a compelling justification for their use.

The standard causal interpretation of 2SLS is rooted in the Local Average Treatment Effect (LATE) framework of Imbens and Angrist (1994). With a single binary instrument alongside a standard monotonicity assumption, 2SLS identifies the average treatment effect for those individuals whose treatment status is changed by the instrument, this kind of individuals are known as the **compliers**. This interpretation has become a cornerstone of modern empirical practice. When multiple instruments are available, Angrist and Imbens (1995) showed that the 2SLS estimand can be interpreted as a positively weighted average of local average treatment effects (LATEs) for subpopulations whose treatment status is affected by the instruments.

However, justifying this interpretation in the multiple-instrument setting requires a strong generalization of the standard monotonicity assumption. As shown by Mogstad, Torgovitsky, and Walters (2021), this assumption is extremely restrictive and is frequently violated once treatment effects are heterogeneous. Importantly, the problem does not arise from high dimensionality or technical complexity. Instead, it is inherent to the use of more than one instrument, even in the simplest possible environments.

To make this point transparent, this paper focuses on the simplest nontrivial setting: a binary treatment and two binary instrumental variables. Even in this minimal environment, the standard monotonicity assumption, which will be called “Imbens and Angrist monotonicity”, or **IAM**, imposes strong restrictions on individual treatment choice behavior. With four possible instrument states, the assumption requires that each individual’s treatment decisions across these states can be ordered in a single, consistent direction. As a result, individuals cannot respond to different instruments in meaningfully different ways.

This restriction is difficult to reconcile with applied work. In empirical practice, different in-

struments typically represent distinct sources of exogenous variation, such as institutional rules, policy thresholds, or geographic constraints. It is therefore natural to expect that different instruments affect different individuals and move different margins of the population into treatment. Nevertheless, such instruments are often combined in a single 2SLS regression and interpreted as if they were reinforcing the same causal effect. The key insight of Mogstad, Torgovitsky, and Walters (2021) is that IAM does much more than rule out defiers: in the presence of multiple instruments, it effectively rules out economically meaningful heterogeneity in how different instruments affect treatment.

The main contribution of the paper is to make this restriction explicit and to propose a more credible alternative. The authors introduce **partial monotonicity**, which requires that each instrument, holding the other instruments fixed, affects treatment in a weakly monotone way. This weaker assumption allows different instruments to influence different subpopulations, while still ruling out perverse responses to each instrument individually.

Under partial monotonicity, 2SLS continues to admit a causal interpretation, but its nature changes in an important way. Rather than identifying a single local average treatment effect, the 2SLS estimand becomes a weighted average of multiple group-specific causal effects, and these weights do not need to be all positive. The paper provides a complete characterization of this estimand, clarifies the economic meaning of the resulting weights, and shows how their signs are directly linked to observable first-stage relationships.

The remainder of this term paper proceeds as follows. Section 2 will provide a detailed overview of Mogstad, Torgovitsky, and Walters' methodology and framework, introduces IAM and partial monotonicity assumptions, and discusses their implications for treatment choice behavior. Section 5 examines the empirical implications of the framework and illustrates the results using a simulation. Section 6 discusses limitations, open questions, and potential extensions, and concludes.

1.1 Literature Review

The starting point for the present paper is the classic LATE interpretation of IV under heterogeneity. Imbens and Angrist (1994) established that with a single binary instrument and monotonicity, IV identifies a complier average causal effect. Angrist and Imbens (1995) then extended this

logic to multiple instruments, showing that 2SLS can be interpreted as a positively weighted average of instrument-specific LATEs under a multi-instrument monotonicity condition. The contribution of the present paper is best understood as a sharp critique of how restrictive this multi-instrument monotonicity requirement becomes.

Related work by the same authors develops complementary implications of weakening global monotonicity. Mogstad, Torgovitsky, and Walters (2024) can be read as a direct extension of the present paper’s perspective from the interpretation of the linear 2SLS estimand to the identification of broader policy-relevant objects. While the present paper characterizes what 2SLS means under partial monotonicity, Mogstad, Torgovitsky, and Walters (2024) uses the same partial monotonicity restriction to extract and aggregate information from multiple instruments for target parameters beyond 2SLS. In this sense, it extends the present paper’s core message, that interpretation depends on behavioral restrictions, into a framework aimed at policy evaluation rather than linear estimation.

Subsequent work has also taken the present paper’s assumptions as objects of empirical scrutiny. Jiang and Sun (2023) addresses a practical implication of the present paper: once one replaces the strong global monotonicity requirement with partial monotonicity, a natural question is whether partial monotonicity is plausible in applications. The paper responds by proposing a simple nonparametric test for partial monotonicity, assessing its finite-sample behavior in simulations, and illustrating it in an empirical application.

A related, more structural refinement is provided by Goff (2024). This paper shares the present Mogstad, Torgovitsky, and Walters (2021) motivation, in that global multi-instrument monotonicity is often too restrictive, but proposes a different weakening, vector monotonicity, which requires treatment uptake to be monotone in each instrument without imposing a global ordering when instruments shift in opposing directions. Relative to the present paper, its contribution is to characterize which causal parameters become point identified under this alternative restriction and to provide constructive estimation results. Conceptually, it highlights that once global monotonicity is relaxed, different monotonicity notions lead to different identifiable objects, reinforcing the present paper’s central emphasis on interpretability rather than mechanical estimation.

2 Methodology Overview

Following the motivation outlined in Section 1, this second chapter contains the methodology developed by Mogstad, Torgovitsky, and Walters (2021).

2.1 Framework and Assumptions

2.1.1 Setup and Potential Outcomes

To fix ideas and keep the exposition transparent, this section focuses on the simplest nontrivial setting considered in the paper: a **binary treatment** and **two binary instrumental variables**. While the theoretical results extend to more general environments, the central insights already arise in this minimal framework.

Let $D_i \in \{0, 1\}$ denote a binary treatment and Y_i the observed outcome for individual i . Each individual is characterized by potential outcomes $Y_i(1)$ and $Y_i(0)$, corresponding to the potential outcomes under treatment and control, respectively. The observed outcome can thus be expressed as:

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0). \quad (1)$$

Treatment assignment depends on two binary instruments,

$$Z_i = (Z_{i1}, Z_{i2}) \in \{0, 1\}^2, \quad (2)$$

which generate four possible instrument realizations: $(0, 0)$, $(0, 1)$, $(1, 0)$, and $(1, 1)$. Treatment choice is described by a potential treatment function $D_i(z_1, z_2)$, allowing for heterogeneous responses across individuals and differential responses to the two instruments.

2.1.2 2SLS Estimand

The object of interest is the two-stage least squares estimand, denoted β_{2SLS} . The population 2SLS estimand can be written as

$$\beta_{2SLS} = \frac{\mathbb{E}[\tilde{D}_i \tilde{Y}_i]}{\mathbb{E}[\tilde{D}_i^2]}, \quad (3)$$

where $\tilde{D}_i = \mathbb{E}[D_i|Z_i] - \mathbb{E}[D_i]$ and $\tilde{Y}_i = \mathbb{E}[Y_i|Z_i] - \mathbb{E}[Y_i]$ denote the residualized treatment and outcome with respect to the instrument space. Equivalently, β_{2SLS} can be expressed as

$$\beta_{2SLS} = \frac{\text{Cov}(\mathbb{E}[D_i|Z_i], \mathbb{E}[Y_i|Z_i])}{\text{Var}(\mathbb{E}[D_i|Z_i])}. \quad (4)$$

In the binary-instrument setting, this representation highlights that the 2SLS estimand aggregates outcome differences across instrument-induced changes in treatment propensity. Under standard instrumental variables assumptions, this estimand is well defined and consistently estimated by sample 2SLS.

The focus of the present paper is not on estimation or consistency, but on the causal interpretation of β_{2SLS} . In particular, the central question is, under which assumptions on treatment choice behavior can β_{2SLS} be written as a weighted average of causal effects, and which causal effects are being aggregated? Addressing this question requires explicit assumptions on how the instruments affect treatment.

2.2 Imbens and Angrist Monotonicity (IAM)

A natural starting point is the extension of the standard monotonicity assumption from the single-instrument LATE framework, known as Imbens and Angrist Monotonicity (IAM).

Definition 1 (Imbens and Angrist Monotonicity)

For all $z, z' \in \mathcal{Z}$ either $D_i(z) \geq D_i(z')$ for all $i \in \mathcal{I}$, or $D_i(z) \leq D_i(z')$ for all $i \in \mathcal{I}$.

Under IAM, there must exist a global ordering of the four instrument values such that, for every individual, treatment status moves weakly in the same direction along this ordering.

When IAM holds, the 2SLS estimand admits a causal interpretation as a weighted average of local average treatment effects. Importantly, IAM is not an assumption imposed separately on each instrument. Rather, it is a joint restriction across all instrument combinations.

While this monotonicity assumption in the single-instrument case simply rules out defiers and is often viewed as a mild behavioral restriction, IAM becomes substantially more demanding when multiple instruments are present. In particular, it places strong constraints on heterogeneity in treatment choice by requiring all instruments to operate along the same margin of individuals.

2.3 Partial Monotonicity

To address the restrictiveness of IAM, the paper proposes a weaker alternative called partial monotonicity.

Definition 2 (Partial Monotonicity)

Take any $l = 1, \dots, L$, and let (z_l, z_{-l}) and (z'_l, z_{-l}) be two points in $\mathcal{Z}$. Then, either $D_i(z_l, z_{-l}) \geq D_i(z'_l, z_{-l})$ for all $i \in \mathcal{I}$, or $D_i(z_l, z_{-l}) \leq D_i(z'_l, z_{-l})$ for all $i \in \mathcal{I}$.

Partial monotonicity relaxes the global ordering requirement imposed by IAM while retaining enough structure to support causal interpretation.

Formally, partial monotonicity requires that each instrument affects treatment in a weakly monotone way when **holding the other instrument fixed**. That is, increasing Z_{i1} from 0 to 1 cannot induce some individuals to enter treatment and others to exit treatment conditional on Z_{i2} , and analogously for Z_{i2} .

Crucially, partial monotonicity does not impose a global ordering across all instrument combinations. As a result, it allows exactly the type of heterogeneity that IAM rules out. Different instruments may affect different subpopulations, one instrument may induce treatment for one group of individuals, while the other operates on a distinct margin. From an economic perspective, this behavior is often natural when instruments capture different incentives or constraints.

2.3.1 Response Groups under Partial Monotonicity

The partial monotonicity assumption implies that individuals can be classified into distinct response groups based on how their treatment status responds to changes in each instrument. In the binary-instrument setting, we have

$$Z_i = (Z_{i1}, Z_{i2}) \in \{0, 1\}^2, \quad (5)$$

which generate four possible instrument values $z_1 = (0, 0)$, $z_2 = (0, 1)$, $z_3 = (1, 0)$, $z_4 = (1, 1)$. Treatment choice is described by the potential treatment function $D_i(z_1, z_2)$, which specifies whether individual i receives treatment under each instrument realization.

The following proposition, establishes that the population can be partitioned into six exclusive

response groups.

Proposition 1¹:

If $\mathcal{Z} = \{0, 1\}^2$ and partial monotonicity holds with the following order:

$$D_i(0, 0) \leq D_i(0, 1) \leq D_i(1, 1), \quad (6)$$

and

$$D_i(0, 0) \leq D_i(1, 0) \leq D_i(1, 1), \quad (7)$$

then each $i \in \mathcal{I}$ belongs exactly to one of the six groups in Table 1.

Group G_i	$D_i(0, 0)$	$D_i(0, 1)$	$D_i(1, 0)$	$D_i(1, 1)$
Always-taker (at)	1	1	1	1
Eager complier (ec)	0	1	1	1
Reluctant complier (rc)	0	0	0	1
Never-taker (nt)	0	0	0	0
Z_1 complier (c1)	0	0	1	1
Z_2 complier (c2)	0	1	0	1

Table 1: Response Groups under Partial Monotonicity

Since the focus is on the causal interpretation of 2SLS, four of these six groups are particularly relevant:

1. **Z_1 compliers (c1):** individuals whose treatment status changes when Z_{i1} switches from 0 to 1, holding Z_{i2} fixed, but who do not respond to changes in Z_{i2} .
2. **Z_2 compliers (c2):** individuals whose treatment status changes when Z_{i2} switches from 0 to 1, holding Z_{i1} fixed, but who do not respond to changes in Z_{i1} .
3. **Eager compliers (ec):** individuals who enter treatment when either instrument is activated. For these individuals, both instruments independently push them into treatment.
4. **Reluctant compliers (rc):** individuals who enter treatment only when both instruments are active. Each instrument alone is insufficient, but their joint activation induces treatment.

¹This proposition is adapted from Mogstad, Torgovitsky, and Walters (2021).

These groups generalize the familiar complier concept from the single-instrument LATE framework and reflect the fact that different instruments may operate on different margins of the population. In particular, individuals are allowed to respond differently to different instruments, as long as each instrument moves treatment in a consistent direction holding the other fixed.

2.4 2SLS Estimand under Partial Monotonicity

2.4.1 Instrumental Variable Assumptions

The 2SLS estimand under partial monotonicity is derived naturally under partial monotonicity, but also under the standard instrumental-variables assumptions.

Instrument Relevance: *Each instrument has a nonzero effect on the probability of treatment:*

$$\mathbb{P}(D_i = 1 | Z_{i,l} = 1) \neq \mathbb{P}(D_i = 1 | Z_{i,l} = 0), \quad l \in \{1, 2\}. \quad (8)$$

Instrument Exogeneity: *The instruments are independent of potential outcomes and potential treatments:*

$$(Y_i(0), Y_i(1), \{D_i(z)\}_{z \in \mathcal{Z}}) \perp Z_i. \quad (9)$$

Exclusion: *Instruments affect the outcome only through treatment:*

$$Y_i(z) = Y_i(D_i(z)) \quad \text{for all } z \in \mathcal{Z}. \quad (10)$$

2.4.2 Causal Interpretation of 2SLS under Partial Monotonicity

Theorem 1²:

*Suppose the instrument vector $Z_i = (Z_{i1}, Z_{i2})$ has support $\mathcal{Z} = \{0, 1\}^2$, and that **partial monotonicity, instrument relevance, instrument exogeneity, and the exclusion restriction** hold. Suppose further that the potential outcomes are well-defined and that the 2SLS estimand β_{2SLS} exists. Then, the 2SLS estimand can be expressed as:*

$$\beta_{2SLS} = \sum_{g \in \{1c, 2c, ec, rc\}} w_g \Delta_g, \quad (11)$$

²This theorem is adapted from Mogstad, Torgovitsky, and Walters (2021).

where Δ_g denotes the average treatment effect for response group g , and w_g are weights with $\sum_g w_g = 1$. Both the weights of eager compliers and reluctant compliers are always nonnegative, i.e. $w_{ec}, w_{rc} \geq 0$, while one of the weights of Z_1 compliers or Z_2 compliers can be negative.

If $\pi_{1c} \geq \pi_{2c}$ where $\pi_g := \mathbb{P}(G_i = g)$, then w_{1c} is also nonnegative, while the sign of w_{2c} is given by:

$$\text{sgn}(w_{2c}) = \mathbb{I}[\pi_{2c} > 0] \times \text{sgn}[\mathbb{P}(D_i = 1|Z_{i,2} = 1) - \mathbb{P}(D_i = 1|Z_{i,2} = 0)]. \quad (12)$$

If $\pi_{2c} \geq \pi_{1c}$, then w_{2c} is nonnegative, while the sign of w_{1c} is given by

$$\text{sgn}(w_{1c}) = \mathbb{I}[\pi_{1c} > 0] \times \text{sgn}[\mathbb{P}(D_i = 1|Z_{i,1} = 1) - \mathbb{P}(D_i = 1|Z_{i,1} = 0)]. \quad (13)$$

This theorem establishes that the 2SLS estimand can be interpreted as a weighted average of average treatment effects for the four possible groups that change treatment status in response to the one or both of the instruments.

The economic intuition is that some changes in the instrument values, specifically the shift from $(0, 1)$ to $(1, 0)$ can generate two-way treatment flows, this means that one group of individuals enters treatment while another exits. When this happens, the associated complier group receives a negative weight.

This result provides a complete characterization of what 2SLS identifies under partial monotonicity. Rather than a single LATE, the estimand aggregates causal effects across multiple response groups, with the sign and magnitude of the weights determined by observable first-stage relationships.

Proof of Theorem 1³:

The goal of the proof is to show that, under partial monotonicity and standard IV assumptions, the 2SLS estimand can be expressed as a weighted average of group-specific treatment effects, and to characterize precisely when one of these weights may be negative.

³This proof is also adapted from Mogstad, Torgovitsky, and Walters (2021).

The proof begins by labeling the four possible realizations of the two binary instruments as

$$z^1 = (0, 0), z^2 = (0, 1), z^3 = (1, 0), z^4 = (1, 1), \quad (14)$$

and denoting their probabilities as $q_k = \mathbb{P}(Z_i = z^k)$ for $k = 1, \dots, 4$. Let

$$p_k = \mathbb{P}[D_i = 1 | Z_i = z^k], \quad (15)$$

denote the propensity score associated with instrument value z^k . Under partial monotonicity and the ordering assumed in the paper, these propensity scores are weakly increasing in k . A key result used in the proof is a result due to Angrist and Imbens, which shows that in this setting the 2SLS estimand can be written as a convex combination of three Wald estimands corresponding to adjacent changes in the instrument values, i.e.

$$\beta_{2SLS} = \lambda_2 w_2 + \lambda_3 w_3 + \lambda_4 w_4, \quad (16)$$

where the coefficients λ_k are positive weights and sum to one. The Wald estimands w_k are defined as

$$w_k := \frac{\mathbb{E}[Y_i | Z_i = z^k] - \mathbb{E}[Y_i | Z_i = z^{k-1}]}{p(z^k) - p(z^{k-1})}, \quad k = 2, 3, 4. \quad (17)$$

Each Wald estimand captures the average causal effect for individuals whose treatment status changes when the instrument moves from z^{k-1} to z^k . The weights λ_k are defined as

$$\lambda_k := \frac{(p(z^k) - p(z^{k-1})) \sum_{l=k}^4 q^l (p(z^l) - \mathbb{E}[p(Z_i)])}{\sum_{j=2}^4 [(p(z^j) - p(z^{j-1})) \sum_{l=j}^4 q^l (p(z^l) - \mathbb{E}[p(Z_i)])]}. \quad (18)$$

The group definitions from Proposition 1 implies that w_2 represents the average treatment effect for both the Z_2 compliers and the eager compliers. Similarly, w_4 captures the average treatment effect for Z_2 compliers and reluctant compliers, i.e.

$$w_2 = \left(\frac{\pi_{2c}}{\pi_{2c} + \pi_{ec}} \right) \Delta_{2c} + \left(\frac{\pi_{ec}}{\pi_{2c} + \pi_{ec}} \right) \Delta_{ec} \quad (19)$$

and

$$w_4 = \left(\frac{\pi_{2c}}{\pi_{2c} + \pi_{rc}} \right) \Delta_{2c} + \left(\frac{\pi_{rc}}{\pi_{2c} + \pi_{rc}} \right) \Delta_{rc}. \quad (20)$$

For the transitions from $z^1 \rightarrow z^2$ and from $z^3 \rightarrow z^4$, only one instrument changes while the other is held fixed. Under partial monotonicity, these transitions induce one-directional treatment movements, i.e. individuals either enter treatment or remain untreated. As a result, the corresponding Wald estimands can be written as weighted averages of treatment effects for well-defined complier groups. Importantly, all weights involved in these decompositions are nonnegative.

However, the crucial part comes from the transition from $z^2 = (0, 1) \rightarrow z^3 = (1, 0)$, which is different. Along this margin, one instrument increases while the other decreases. Partial monotonicity allows this change to generate **two-way treatment flows**, where Z_1 compliers enter treatment while Z_2 compliers exit treatment. Consequently, the Wald estimand w_3 can be expressed as

$$\begin{aligned} w_3 &= \frac{\mathbb{E}[(Y_i(1) - Y_i(0))(D_i(1, 0) - D_i(0, 1))]}{p(z^3) - p(z^2)} \\ &= \left(\frac{\pi_{1c}}{p(z^3) - p(z^2)} \right) \Delta_{1c} + \left(\frac{\pi_{2c}}{p(z^3) - p(z^2)} \right) \Delta_{2c} \\ &= \left(\frac{\pi_{1c}}{\pi_{1c} - \pi_{2c}} \right) \Delta_{1c} + \left(\frac{\pi_{2c}}{\pi_{1c} - \pi_{2c}} \right) \Delta_{2c}. \end{aligned} \quad (21)$$

Substituting the decompositions of the three Wald estimands back into the convex combination for β_{2SLS} yields

$$\beta_{2SLS} = \sum_{g \in \{1c, 2c, ec, rc\}} w_g \Delta_g, \quad (22)$$

where

$$\begin{aligned} w_{ec} &:= \frac{\lambda_2 \pi_{ec}}{\pi_{ec} + \pi_{2c}}, \\ w_{1c} &:= \frac{\lambda_3 \pi_{1c}}{\pi_{1c} - \pi_{2c}}, \\ w_{rc} &:= \frac{\lambda_4 \pi_{rc}}{\pi_{rc} + \pi_{2c}}, \\ w_{2c} &:= \left(\frac{\lambda_2 \pi_{2c}}{\pi_{ec} + \pi_{2c}} \right) - \left(\frac{\lambda_3 \pi_{2c}}{\pi_{1c} - \pi_{2c}} \right) + \left(\frac{\lambda_4 \pi_{2c}}{\pi_{rc} + \pi_{2c}} \right). \end{aligned}$$

The proof shows that the weights on eager and reluctant compliers are always nonnegative, and that at most one instrument-specific complier group can receive a negative weight. Moreover,

the sign of this potentially negative weight is not arbitrary, it is entirely determined by an observable first-stage object, specifically by the covariance between treatment and the corresponding instrument. In particular, the weight on Z_2 compliers is negative if and only if the first-stage effect of Z_2 on treatment is negative along the relevant margin.

The central insight of the proof is therefore that negative weights arise only when an instrument change induces offsetting treatment flows, i.e. some individuals enter treatment while others exit. Apart from this specific case, all components of the 2SLS estimand receive nonnegative weight.

3 Simulation Study

4 Discussion

4.1 Summary

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